

Subject Description Form

Subject Code	EIE2112
Subject Title	Foundation Techniques in Artificial Intelligence
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To introduce the latest development of Artificial Intelligence (AI) and Information Engineering (IE). 2. To introduce the common mathematical and programming tools used in the study of Artificial intelligence and Information Engineering.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the relationship between AI, machine learning and applications; 2. Understand the common mathematical tools used in the study of AI and IE; 3. Apply computer programming techniques to solve practical scientific problems; and <u>Category B: Attributes for all-roundedness</u> 4. Solve problems independently.
Subject Synopsis/ Indicative Syllabus	1. <u>Introduction to AI and IE</u> • Introduction to Artificial Intelligence and Machine Learning • Techniques and technologies in Artificial Intelligence and Information Engineering: Natural language processing, Linear regression, Decision tree, Support Vector Machine, Neural Networks, Clustering, Dimensionality reduction 2. <u>Mathematical Foundations of AI and IE</u> • Complex number: geometric representation, complex exponential functions, n-th roots of a complex number • Linear algebra: Review of basic matrix operations. Determinants and systems of linear equations. Inner product and orthogonality, eigenvalues and eigenvectors. Case study: Real life application of linear algebra. 3. <u>Machine Learning</u> • Vectors, matrices, tensors, and vector space • Random variables and probability distributions • Bayes theorem and its applications • Supervised learning versus unsupervised learning • Overfitting, underfitting, and dimension reduction • Gaussian mixture models and Support vector machines

Teaching/Learning Methodology	Teaching and Learning Method	Intended Subject Learning Outcome	Remarks			
	Lectures	1,2,3	Fundamental principles and key concepts of the subject are delivered to students.			
	Tutorials	1,2,3	Supplementary to lectures: Students will be able to clarify concepts and to have a deeper understanding of the lecture materials; Problems and applications are given and discussed.			
	Laboratory sessions	2,3,4	Students will experience the applications of different mathematical tools by means of some computer programming experiments in numerical computation.			

Assessment Methods in Alignment with Intended Subject Learning Outcomes	Specific Assessment Methods/Tasks		% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)			
				1	2	3	4
	1. Short quizzes		10%	✓	✓	✓	
	2. Laboratory sessions		30%		✓	✓	✓
	3. Examination		60%	✓	✓	✓	✓
	Total		100%				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:						
	Specific Assessment Methods/Tasks		Remark				
	Short quizzes		They can measure the students’ understanding of the theories and concepts as well as their comprehension of subject materials.				
	Examination		End-of-chapter-type problems are used to evaluate the students’ understanding of subject materials and the ability in applying concepts and skills learned in the classroom. Students need to think critically and to learn independently in order to come up with an alternative solution to an existing problem. They need to present their solutions logically and systematically in the tests and the examination.				
Laboratory sessions		Students are required to make a demonstration of their solutions on a selected open-ended question in each laboratory session for evaluating their problem solving skill. Students also need to submit lab reports for evaluating their overall performance in the laboratory sessions.					

Student Study Effort Expected	Class contact (time-tabled):	
	Lectures	24 Hours
	Tutorial/Laboratory/Practice Classes	15 Hours
	Other student study effort:	
	Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes	36 Hours
	Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	30 Hours
	Total student study effort:	105 Hours
Reading List and References	1. M.J. Roberts, <i>Fundamentals of Signals & Systems</i>, McGraw-Hill, 2008. 2. R. Larson, Edwards, B. <i>Single Variable Calculus</i>, Brooks/Cole 2012 3. R. Larson, <i>Elementary Linear Algebra</i>, Brooks/Cole 2013 4. S. Nagar, <i>Introduction to Python for Engineers and Scientists: Open Source Solutions for Numerical Computation</i>, Apress, 2018 5. Joshi Ameet, “<i>Machine learning and artificial intelligence</i>”, Springer 2020. 6. Ashish Ranjan Jha, Dr. Gopinath Pilai, “<i>Mastering PyTorch</i>”, Packt Publishing 2021. 7. Singh Pramod and Manure Avinash, “<i>Learn TensorFlow 2.0: Implement Machine Learning and Deep Learning</i>”, Apress, 2020.	

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